INCH-POUND

MIL-DTL-24643/65
1 March 2007

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3C179DT
(INCLUDING VARIATION LS6C179DT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS3C179DT (three internal coaxial cables) and LS6C179DT (six internal coaxial cables).

REQUIREMENTS:

Qualification Required.

Construction (Non-Watertight)

| | | |
|---|---|---|
| First | - | Copper conductor, uncoated, in accordance with ASTM B286, 0.0122-inch diameter. |
| Second | - | Thermoset insulation, diameter 0.056 inch nominal. |
| Third | - | Light bonded aluminum foil-polyester tape, aluminum foil shield. Foil in contact with braid. |
| Fourth | - | Braided shield of 38 AWG tin-coated copper: minimum coverage 95 percent, diameter 0.079 inch nominal. |
| Fifth | - | Cross-linked polyolefin jacket. Diameter 0.100±0.005 inch. |
| Sixth | - | The required number of coaxial cables cabled together with a lay in accordance with 3.4.9 of MIL-DTL-24643. Fillers shall be used as necessary to make a firm, well-rounded assembly. |
| Seventh | - | An optional binder tape applied helically with overlap. |
| Eighth | - | Tinned copper braid, 36 gage, 90 percent coverage. |
| Ninth | - | Cross-linked polyolefin jacket. Minimum thickness of 0.040 inch. |



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TABLE I. Details.

| Military Part No. M24643/XX | Size | Number of Coaxes | Diameter over Shield (inches nominal) | Cable Jacket Thickness Min. Avg. (inches) | Overall Diameter (inches) | | Cold Bend Mandrel Diameter (inches) |
|---|---|---|---|---|---|---|---|
| | | | | | Min. | Max. | |
| -01UO | 3 | 3 | 0.240 | 0.047 | 0.335 | 0.365 | 4.0 |
| -02UO | 6 | 6 | 0.325 | 0.050 | 0.425 | 0.455 | 5.0 |

EXAMINATION AND TESTS:

| | |
|---|---|
| Basic Electricals: | |
| Conductor resistance (ohms/1000 feet at 25 °C, max.) | 110 |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor | 2000 |
| Conductor to shield | 1000 |
| Insulation resistance (megohms/1000 feet, min.) | |
| Conductor to conductor | 500 |
| Conductor to shield | 500 |
| Jacket flaws | No Failure |
| Conductor and shield continuity | No Failure |
| Group A: | |
| Visual and dimensional | No Failure |
| Characteristic impedance (ohms) (TDR method in accordance with characteristic impedance of MIL-DTL-17) | 75±5 |
| Group B: | |
| Thermoset proof test (percent, max.) | |
| Overall jacket | 50 |
| Cold bending, cable mandrel diameter 12 times cable diameter | No Damage |
| Bending endurance – at 25+2 °C (cycles, min.) | 25 |
| Tear (pounds per inch thickness, min.) (ASTM D470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Tensile strength ($lb/in^2$, min.) | 600 |
| Elongation (percent, min.) | 200 |
| Jacket (component) | |



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|---|---|
| Tensile strength (lb/in$^2$, min.) | 1000 |
| Elongation (percent, min.) | 100 |
| Jacket (cable) | |
| Tensile strength (lb/in$^2$, min.) | 1300 |
| Elongation (percent, min.) | 160 |
| Group C: | |
| Physicals (aged) | |
| Insulation | |
| Air oven | |
| Tensile strength (percent of unaged, min.) | 80 |
| Elongation (percent of unaged, min.) | 80 |
| Jacket (components) | |
| Air oven | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Jacket (cable) | |
| Air oven | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Shrinkage | No Failure |
| Heat distortion (percent of unaged, max.) | 30 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Group D: | |
| Flame propagation (cable) | No Failure |
| Qualification Inspection: | |
| Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following: | |
| Aging stability (cable) (98±2 °C)<br>(Test in accordance with aging stability of MIL-DTL-17) | No Failure |
| Dimensional stability (cable) (85±2 °C)<br>(Test in accordance with dimensional stability of MIL-DTL-17) | No Failure |
| Abrasion resistance (overall jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket (component and overall) | 2 |



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|---|---|
| Fillers | 2 |
| Insulation | 18 |
| Shield insulation | 2 |
| Halogen content (percent, max.) | |
| Jacket (component and overall) | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index, max. | |
| Jacket (component and overall) | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index, max. | |
| Jacket (component and overall) | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (overall jacket) – Type A (hardness, min.) | 80 |
| Weathering (overall jacket) | No Failure |

UNIT ORDERING LENGTHS: All sizes 500 feet (nominal).

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2006-069)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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